LastOne

Product Foundations Document

The rules of the world this product lives in.

Part A: Team Reference | Part B: System Logic
Version 1.0 — March 2026

How to Use This Document

This document defines the five foundational concepts that everything in LastOne is built on. It is not a feature list. It is not a design spec. It is the set of structural decisions that every feature, screen, and interaction must follow.

Part A is for the design team. Read it once fully, then use it as a reference when making design decisions. It answers: what are the rules, and why do they exist?

Part B is for backend implementation (Claude Code or a developer). It translates the same concepts into explicit logic: variables, conditions, thresholds, and data structures.

The five foundations, in order:

- **1. Stage System** — the quitting timeline that determines when everything happens
- **2. Two-Layer Coping System** — how tools are surfaced during and between cravings
- **3. Social Architecture** — how users connect with friends, nearby quitters, and community
- **4. Personalisation Model** — what data is collected and how it adapts the experience
- **5. Content & Awareness Framework** — what content exists, how it's delivered, and in what format

> *V2 items documented elsewhere: ANT (Ambient Narrative Interface) and the Supporter Role. These are parked for the current version.*

PART A: TEAM REFERENCE

1. The Stage System

Quitting isn't one thing — it's a series of different challenges that show up at different times. The stage system is the backbone of LastOne. It determines when content gets shown, which coping tools surface, how notifications behave, and what counts as progress.

> *Think of it as LastOne's calendar. Every feature checks what stage the user is in before deciding what to do.*

Stage 0 — The Learning Week (Days -7 to 0)

The user hasn't quit yet. They smoke normally while LastOne learns their patterns. They log every cigarette (one-tap + optional context: where, with whom, trigger, mood). The app builds their smoking profile — timing, triggers, social patterns. Mid-week and end-of-week, the app reflects their data back to them. Light awareness content drips in. The quit date is set at the end of this week.

> *The user must feel that logging is worthwhile. The app gives back small insights during this week, not just after it.*

Stage 1 — First 72 Hours (Days 1–3)

Peak physical withdrawal. Maximum support intensity. The curated coping layer is most active. Notifications run 2–3 per day, timed to the user's known high-risk windows from the Learning Week. Content focuses on withdrawal normalisation and early health milestones (heart rate, blood CO). The app's tone is honest, grounding, never artificially cheerful.

Stage 2 — First Full Week (Days 4–7)

Physical withdrawal is fading but habitual triggers are hitting hard. This is the first Monday, the first exam prep, the first social situation without a cigarette. The app shifts from crisis mode to pattern-based coaching — using Learning Week data to preempt trigger moments. Myth-busting and Allen Carr-style reframing content starts here. Notifications taper to 1–2 per day.

Stage 3 — Weeks 2–3 (Days 8–21)

Physical withdrawal is essentially over. The challenge is now psychological and social — old cues still fire, but the body isn't demanding nicotine. Habit disruption and social features become most important. The insights engine shows the user their declining craving trends. Gamification milestones mark progress. Notifications taper further.

Stage 4 — Weeks 4–8 (Days 22–56)

Stabilisation. The user has made it through the acute phase. The threat is now complacency — feeling fine and letting guard down. Weekly summaries replace daily check-ins. Money saved hits tangible amounts. The community and social layer become primary engagement drivers. Notifications drop to 2–3 per week.

Stage 5 — Months 3+ (Day 57 onwards)

Long-term maintenance. Low-frequency, high-value touchpoints. Monthly check-ins. Major milestone celebrations. The health recovery timeline continues unlocking. The user shifts from needing support to giving it — becoming a community contributor.

Relapse Rules

The core principle: setbacks reduce momentum but never erase progress. The user's data, insights, and tool preferences always persist across restarts. A restart resets the quit attempt timeline, not the relationship with the app.

Stage	Allowance	What Happens
Stage 1 (Days 1–3)	1–2 slips absorbed	Log it, reflect, continue. User can choose to restart Stage 1.
Stage 2 (Days 4–7)	1 slip absorbed	3+ slips → app gently surfaces restart option. Restart = short 2–3 day prep + Stage 1.
Stage 3 (Days 8–21)	1–2 slips absorbed	3+ slips in 14 days → app recommends restart. Same short prep + Stage 1.
Stage 4 (Days 22–56)	1–2 slips absorbed	3+ in a 2-week window → firmer restart recommendation.
Stage 5 (Months 3+)	Isolated slips absorbed	Sustained pattern (4–5 over weeks) → soft recommendation for a focused reset.

> *Language direction: consider framing restarts as 'iterations' not 'resets.' The user is on Version 2 of their quit attempt, built on what Version 1 taught them. Not starting over — starting smarter.*

2. The Two-Layer Coping System

A craving lasts 3–5 minutes. The coping system's job is to help the user survive those minutes. Same pool of tools, two different surfaces depending on the moment.

Layer 1: Curated Surface (during active cravings)

When the user taps 'I'm craving' or the app proactively reaches out during a known high-risk time, they see 2–3 tools maximum. No browsing, no choosing from a long list. The app picks for them based on: their current stage, what's worked for them before, and the craving context (if available). Under 5 seconds from 'I'm craving' to 'I'm doing something about it.'

Layer 2: Explorative Library (between cravings)

The full tool pool, browsable by category. Accessible from a dedicated section. Users explore at their own pace — trying tools before they need them, building familiarity. Each tool shows a personal effectiveness marker after they've used it. Daily-use engagement tools (games, puzzles) also live here.

What's in the Coping Pool

The tools fall into categories, but individual tools are guided micro-experiences that blend physical action with cognitive prompts. A reframing prompt never appears as standalone text — it's always embedded in a 30–60 second guided sequence (breathe → notice → reflect → reframe).

Category	What It Contains	When It's Most Relevant
Physical Reset	Short breathing patterns (30–60 sec), movement prompts, muscle tension-release, hydration prompt	Stages 1–2 (withdrawal is dominant)
Cognitive Reframing	Guided micro-experiences that pair breathing/grounding with reflective prompts. Never standalone text.	Stages 2–3 (psychological triggers)
Distraction / Engagement	Quick-play games, puzzles, fidget interactions. Must be genuinely engaging, not wellness-flavoured.	All stages; doubles as boredom/engagement mechanic
Social Coping	SOS tap to quit group, see friends' activity, browse community coping stories	Stages 2–5 (social triggers)
Reflective	Quick mood check-in, post-craving reflection, micro-journaling prompts. Embedded in flow, never standalone.	Stages 2–4 (building self-awareness)

How the System Learns

Each craving → tool used → outcome (resisted or slipped) is logged. Tools get a simple score: +1 for successful resistance, -1 for a slip. The curated layer surfaces the highest-scoring tools for that user's context. Early on, defaults are based on onboarding data. Over time, observed effectiveness takes over.

> *Framing principle: coping tools are NOT replacements for cigarettes. They are things to do while the craving passes on its own. The craving will pass whether they use a tool or not. The tool makes the waiting easier.*

3. Social Architecture

Smoking is a social behaviour. Quitting in isolation while your friends keep smoking is one of the primary reasons quit attempts fail. The social layer exists to keep quitting from being lonely. Three layers, all in-app.

Layer 1: Quit Groups (existing friends)

Small groups of 2–6 people the user already knows. Created by invite link. Each member's profile shows their stage, streak, and whatever they've chosen to share. The group is a shared dashboard with an optional chat layer. When someone slips, their streak visual changes subtly — but the detail stays private unless they choose to share it. No surveillance. Nudged most during Stage 0 and Stage 2.

Layer 2: Nearby Quitters (discovery)

Opt-in discovery of other LastOne users in the same area. Three privacy tiers: fully visible (name, photo, stage, location radius), semi-visible (pseudonym, avatar, broad area), or hidden (browse others without appearing). Connection requires mutual acceptance. The app never shows exact location. Meeting up is decided between two people after connecting through messaging.

Layer 3: Community Feed

A shared space for journey posts, questions, and encouragement. Users can post under their name or anonymously (per-post toggle). Moderated by the LastOne team. Reporting tools for all users. Becomes most relevant from Stage 3 onwards, when the user has bandwidth to engage with others' stories and eventually contribute their own.

Stage Connection

- **Stage 0:** Nudge Quit Group creation. Social setup while motivation is fresh.
- **Stages 1–2:** Quit Groups most active. SOS coping tool pings the group. Nearby Quitters gets a gentle nudge.
- **Stage 3:** Community Feed becomes a useful engagement surface. Nearby discovery more relevant.
- **Stages 4–5:** Community becomes primary social engagement. User shifts from needing support to giving it.

4. Personalisation Model

Personalisation in LastOne is invisible. The user never sees 'Your Profile: Social Smoker.' They just feel 'this app gets me.' Two data sources power it: what the user tells us (onboarding) and what we observe (behaviour).

Onboarding — Declared Data

Five chapters, each serving a distinct purpose:

- **Chapter 1 — Who are you?** Name, age, occupation, location. For social layer and cultural relevance.
- **Chapter 2 — What does your smoking look like?** Smoking days in last 30, cigarettes per smoking day, time to first cigarette + urge strength. Sorts user into one of three profiles (see below).
- **Chapter 3 — Why do you smoke?** Multi-select triggers (friends, stress, boredom, meals, breaks). Previous quit attempt history. Primary motivation for quitting.
- **Chapter 4 — How do you want to be supported?** Voice style preference (Steady & Direct / Warm & Grounding / Light & Honest) — shown as examples, not labels. Notification frequency preference.
- **Chapter 5 — Set your quit date.** Approximately 1 week out. Marks the start of the Learning Week.

The Three User Profiles

Profile	Who They Are	How the App Adapts
Social / Occasional	Non-daily, low dependence, smokes mostly with friends	Lighter notifications. Content weighted toward social pressure and identity. Social layer nudged early.
Regular Light	Daily, <10 CPD, low-moderate dependence	Moderate notifications. Balanced content. Strong emphasis on Learning Week data and habit disruption.
Regular Moderate-to-Heavy	Daily, 10+ CPD or high dependence	Higher notifications in Stages 1–2. Leads with withdrawal normalisation. Closer engagement monitoring.

> *These are starting positions. As observed data accumulates, actual behaviour overrides the initial profile. A social smoker with surprisingly intense cravings will see support intensity adapt upward.*

Learning Week + Ongoing — Observed Data

During the Learning Week: smoking timestamps, context tags, post-cigarette reflections. After quitting: craving logs, tool effectiveness, engagement patterns, craving trend data. This observed data is what makes the curated coping layer and content delivery smarter over time.

What Personalisation Changes

- **Coping tools:** Which tools the curated layer surfaces, based on profile + observed effectiveness.
- **Content:** Which content types get emphasis (money for money-motivated users, health for health-motivated).
- **Notifications:** Timing (matched to personal risk windows), frequency (preference × stage rules), and voice style.
- **Insights:** Which metrics lead weekly summaries (social craving trends for social smokers, frequency trends for daily smokers).

- **Tone:** All high-sensitivity copy adapts to the user's chosen voice style. Switchable anytime from settings.

5. Content & Awareness Framework

Content in LastOne is not a section of the app. It is a layer that runs through everything. There is no 'Learn' tab. Content is woven into tracking, coping, progress, and social features. Every piece appears because the user's stage, data, or behaviour makes it relevant right now.

Five Content Types

- **Health Recovery Milestones —** Body recovery timeline tied to the stage system. Unlocks in real time. Visual, celebratory, gain-framed.
- **Myth-Busting & Reframing —** Allen Carr-adjacent insights. Surfaced through the user's own data wherever possible, not as lectures. Interactive format (reveal, quiz, before/after). Never preachy.
- **Motivational & Reflective Prompts —** Short, personal, moment-specific. Appear after craving logs, in weekly summaries, at milestones. The user reflects, not studies. One sentence each.
- **Practical Tips —** Actionable advice for specific situations. Culturally and situationally specific. Tied to the user's known triggers and timed to their risk windows.
- **Progress & Gain-Frame —** Money saved (in rupees + relatable purchases), time reclaimed, health recovery. Always visible on home screen and widget. Always updating.

Delivery Channels

Content reaches the user through four channels:

- **Home screen carousel —** 3–5 curated cards, refreshed daily by stage + personalisation. Not the full library. Always relevant to today. Tapping expands. Different content types expand differently (visual for milestones, interactive for myths, response-space for reflections).
- **Notifications —** The hook, not the full content. Short, emotionally engaging, tapping opens the relevant card in-app. Written in the user's chosen voice style.
- **Contextual moments —** Content embedded in other features: post-craving, post-milestone, within weekly summaries. The user encounters it naturally, not by seeking it.
- **Personal deck —** The full collection of cards the user has accumulated. Browsable, filterable by type (colour-coded). Locked future cards visible with stage indicators. A personal collection, not a library.

Delivery Rules

- **No education section.** Content lives inside features, never in its own tab.
- **Under 30 seconds per piece.** If it's longer, it's too long for this audience.
- **Every piece has a trigger condition.** Nothing is random. Stage, data, behaviour, or time-of-day makes it relevant.
- **Gain-frame everything.** Not 'smoking causes X' but 'by not smoking, you're gaining X.'
- **User's own data is the strongest content.** Their patterns, their money, their triggers. Personalised > generic.

Voice Styles (for high-sensitivity content only)

Three voices, applied to notifications, craving prompts, milestone celebrations, slip responses, and guided micro-experiences. Not applied to factual content (milestones, tips, stats) — those are written once in neutral-warm tone.

- **Steady & Direct —** Short sentences. Confident. No fluff. 'The craving is here. It'll pass. You know what to do.'
- **Warm & Grounding —** Empathetic, acknowledges difficulty. 'This is a hard moment, and that's okay. Take a breath.'
- **Light & Honest —** Touch of humour, self-aware. 'Your brain is being dramatic. Give it 3 minutes and it'll get bored.'

PART B: SYSTEM LOGIC REFERENCE

This section translates the foundational concepts into implementable logic. Variables, conditions, thresholds, and data structures. Intended for Claude Code or a backend developer.

B1. Stage System Logic

Stage Definitions

Stage	ID	Duration	Entry Condition	Exit Condition
Learning Week	0	7 days (fixed)	Account created + quit date set	Quit date reached
First 72 Hours	1	3 days	Quit date begins OR restart triggered	72 hours elapsed since stage entry
First Full Week	2	4 days	Stage 1 complete	4 days elapsed since stage entry
Weeks 2–3	3	14 days	Stage 2 complete	14 days elapsed since stage entry
Weeks 4–8	4	35 days	Stage 3 complete	35 days elapsed since stage entry
Maintenance	5	Ongoing	Stage 4 complete	None (permanent)

Relapse Logic

Variables:

- slip_count: integer, count of cigarettes logged in current stage window
- stage_window: the duration of the current stage
- rolling_14d_slips: count of cigarettes in last 14 calendar days (used in Stages 3–5)

Stage	Threshold for Restart Suggestion	Restart Target	Data Reset
1	User-initiated only (optional after any slip)	Stage 1 (no prep needed)	None
2	slip_count >= 3 in stage window	2–3 day mini-prep → Stage 1	None
3	rolling_14d_slips >= 3	2–3 day mini-prep → Stage 1	None
4	rolling_14d_slips >= 3	2–3 day mini-prep → Stage 1	None
5	rolling_14d_slips >= 4 OR sustained pattern detected	Focused 2-week reset (Stage 3 equivalent)	None

On any restart: user retains all historical data (logs, insights, tool scores, trigger patterns). quit_attempt_number increments. Previous attempt data is accessible for comparison and personalisation improvement.

B2. Coping System Logic

Tool Selection — Curated Layer

When the curated surface is triggered (user taps 'I'm craving' or proactive push fires), select 2–3 tools using this priority:

- **Priority 1 — Personal effectiveness.** Select tools with highest tool_score for this user. tool_score = (successful_resistance_count - slip_after_use_count). Minimum 3 uses before score is weighted.
- **Priority 2 — Stage weighting.** If insufficient personal data (<3 uses per tool), weight by stage: Stage 1–2 → physical_reset + cognitive_reframe. Stage 3 → distraction + social_coping. Stage 4–5 → social_coping + reflective.
- **Priority 3 — Context matching.** If craving context is provided (stress/boredom/social/habitual), boost tools tagged for that context. If time-of-day matches a known trigger window from Learning Week, boost tools previously successful at that time.
- **Priority 4 — Profile defaults.** If no personal data and no context, use onboarding profile. Social/Occasional → social_coping + cognitive_reframe. Regular Light → physical_reset + distraction. Regular Moderate-Heavy → physical_reset + cognitive_reframe.

Tool Data Structure

Each tool in the pool:

- tool_id: unique identifier
- category: physical_reset | cognitive_reframe | distraction | social_coping | reflective
- context_tags: [stress, boredom, social, habitual, morning, evening, post_meal]
- duration_seconds: integer (target: 30–180)
- stage_relevance: [array of stage IDs where most relevant]
- tool_score: per-user, starts at 0, updated after each craving outcome

B3. Social Architecture Logic

Quit Groups

- max_members: 6
- min_members: 2
- creation: any user, any stage. Invite via shareable link.
- visibility per member: controlled individually. Options: show_stage, show_streak, show_milestones (each boolean, default true)
- slip visibility: streak_visual changes state (paused/dimmed) when slip logged. Detail (count, time, context) remains private unless member posts to group.
- SOS integration: when user triggers SOS from coping system, group members receive a push notification if opted in.

Nearby Quitters Discovery

- opt_in: boolean, default false
- visibility_tier: full | semi | hidden
- discovery_radius: college | city | national (user-selected)
- connection: requires mutual acceptance (request → accept flow)
- messaging: enabled only after mutual connection
- location_precision: broad area label only (e.g. 'Delhi'), never coordinates

Community Feed

- post_identity: per-post toggle (named | anonymous)
- moderation: report_flag per post, auto-filter for blocked terms, moderation queue for flagged posts
- post_types: milestone, reflection, question, encouragement
- feed_algorithm: chronological for V1 (no algorithmic ranking)

B4. Personalisation Logic

Onboarding Data Collection

Field	Input Type	Used For
name	text	Display, social layer
age	number	Content relevance
occupation	select (student/working/other)	Content framing
location	text (city/college)	Nearby discovery, cultural references
smoking_days_30	number (1–30)	Profile classification
cpd_typical	number	Profile classification, money-saved calc
time_to_first_cig	select (within 5min / 6–30min / 31–60min / 60min+)	Dependence proxy
urge_if_cant_smoke	select (strong / moderate / mild / none)	Dependence proxy
triggers	multi-select	Context matching, content priority
previous_quit_attempts	select (0 / 1–2 / 3+)	Messaging, prep customisation
previous_failure_reason	multi-select (if attempts > 0)	Feature emphasis
primary_motivation	multi-select	Gain-frame content priority
voice_style	select (steady / warm / light)	Copy variant selection
notification_preference	select (app_decides / few_daily / once_daily / on_demand)	Notification cadence base
quit_date	date	Stage 0 → Stage 1 transition trigger
price_per_cigarette	number (INR)	Money-saved calculator

Profile Classification Logic

Based on smoking_days_30, cpd_typical, and dependence proxy:

- **Social/Occasional:** smoking_days_30 <= 15 AND cpd_typical <= 5 AND dependence = low
- **Regular Light:** smoking_days_30 >= 16 AND cpd_typical <= 10 AND dependence = low-moderate
- **Regular Moderate-Heavy:** cpd_typical > 10 OR dependence = high

Dependence scoring: time_to_first_cig (within 5min = 3, 6–30min = 2, 31–60min = 1, 60min+ = 0) + urge_strength (strong = 3, moderate = 2, mild = 1, none = 0). Total 0–2 = low, 3–4 = moderate, 5–6 = high.

Notification Cadence Rules

Stage	Base Cadence (app_decides)	Modification Rules
0	1/day	Reminder to log if no log by evening

1	2–3/day	Timed to user's top 3 trigger windows from Learning Week
2	1–2/day	Timed to trigger windows. Preemptive before known risk times.
3	1/day or every other day	Extra push only if risk pattern detected
4	2–3/week	Weekly summary + milestone celebrations
5	1/week or less	Monthly check-in + major milestones

Auto-reduce rule: if user ignores 3 consecutive notifications (no tap), reduce frequency by one tier for 7 days, then test again.

B5. Content System Logic

Content Item Data Structure

Each content item in the library:

- content_id: unique identifier
- type: health_milestone | myth_bust | reflective_prompt | practical_tip | gain_frame
- delivery_stages: [array of stage IDs where this content can surface]
- trigger_condition: stage_entry | time_of_day | data_pattern | user_action | craving_context
- trigger_detail: specific condition (e.g. 'stage_entry:2' or 'craving_context:stress' or 'data:stress_myth_eligible')
- format: visual | interactive | text_only | response_required
- tone_sensitivity: high | low (determines whether voice variants are needed)
- voice_variants: { steady: string, warm: string, light: string } (only if tone_sensitivity = high)
- priority: 1–5 (for carousel slot competition; 1 = highest)
- unlock_condition: stage >= X (for personal deck locked/unlocked state)

Home Carousel Selection Logic

On each app open, select 3–5 cards:

- **Filter:** delivery_stages includes current_stage AND unlock_condition met
- **Sort:** By priority (ascending), then by trigger relevance (content whose trigger_condition matches current context scores higher)
- **Deduplicate:** Max 1 card per type in any single carousel load
- **Freshness:** Content shown in last 24 hours is deprioritised unless priority = 1
- **Personalisation:** Gain-frame content matching user's primary_motivation gets +1 priority boost

Content Production Scope (V1)

Content Type	Approximate Count	Voice Variants Needed
Health recovery milestones	15–20 cards	No (low tone-sensitivity)
Myth-busting cards	10–15 cards	No
Reflective prompts	30–40 prompts	No (short, universal tone)
Practical tips	20–30 tips	No
Gain-frame calculations	20–30 relatable purchase comparisons	No
High-sensitivity copy (notifications, craving prompts, milestones, slip responses)	30–50 unique messages	Yes — each in 3 voice variants (90–150 total)

End of Document

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